

Rahul Lohra

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EDUCATION

Banaras Hindu University (BHU)

Master of Computer Applications (MCA) – CGPA: 7.3/10

Varanasi, India

Sept 2023 – June 2025

Dr. Shyama Prasad Mukherjee University

Bachelor of Science in Information Technology – CGPA: 8.61/10

Ranchi, India

June 2020 – Aug 2023

EXPERIENCE

Software Engineering Intern

Feb 2025 – Aug 2025

CERT-In (Indian Computer Emergency Response Team)

Delhi, India

- Architected and deployed a full-stack Django web application serving 100+ internal users, improving organizational workflow efficiency by 40% through automated document processing and role-based access control
- Implemented enterprise-grade security architecture using PBKDF2-SHA256 password hashing, multi-factor authentication with OTP validation, and Django Axes for brute-force protection, achieving zero security incidents
- Designed and integrated RESTful APIs with Twilio and Gmail services for real-time authentication workflows and automated email notifications, reducing manual verification time by 60%
- Collaborated with cross-functional teams of 8+ members using Agile methodology, participating in daily standups and code reviews to ensure high-quality deliverables within sprint deadlines

PROJECTS

AI-Powered File Converter & Summarizer | *React.js, Django, Flask, NLP, REST APIs* Nov 2024 – Feb 2025

- Built an intelligent document processing platform with React.js frontend and Django/Flask backend, enabling users to convert multiple file formats and generate AI-powered summaries from PDF documents
- Engineered robust backend API endpoints for asynchronous file processing, implementing error handling and request validation to ensure 99% uptime and seamless user experience
- Integrated state-of-the-art NLP models for text extraction and summarization, processing documents 5x faster than manual review while maintaining 90%+ accuracy in content extraction
- Developed responsive frontend with modern React patterns (hooks, context API) and implemented real-time progress tracking for long-running summarization tasks

Federated Learning System | *Python, Scikit-learn, Machine Learning, Data Privacy* Mar 2024 – Aug 2024

- Simulated privacy-preserving Federated Learning framework on medical insurance dataset with 5 distributed clients, demonstrating decentralized model training without raw data sharing
- Implemented SGDR regressor with comprehensive data preprocessing pipeline (normalization, feature engineering) and custom weight aggregation algorithm achieving 85% prediction accuracy
- Developed modular Python-based ML pipeline with comprehensive documentation and unit tests, enabling reproducible experiments and facilitating team collaboration across research phases

TECHNICAL SKILLS

Programming Languages: Python (Advanced), JavaScript (Advanced), SQL, HTML/CSS

Frameworks & Libraries: React.js, Node.js, Express.js, Django, Flask, NumPy, Pandas, Scikit-learn, Matplotlib

Technologies & Tools: Git/GitHub, REST APIs, PostgreSQL, MongoDB, Docker, Jupyter Notebooks, Figma, Agile/Scrum

Core Competencies: Full-Stack Development, Machine Learning, API Design, Data Analysis, System Architecture, Security Implementation, Technical Documentation, Team Collaboration